

The Bungee Jumper Dynamics Model: 1D Simulation and Calculation

April 8, 2026

1 Introduction

The physics of bungee jumping is centered on the balance between gravitational potential energy and elastic restorative forces. Unlike simplified kinematic models that assume constant acceleration, a realistic simulation requires a dynamic approach to handle the transition from free-fall to the variable tension of an expanding cord. This project explores the one-dimensional motion of a jumper through numerical integration, specifically focusing on the transition point where the cord surpasses its natural length and begins to exert a force governed by Hooke's Law.

The primary objective of this project is to calculate the maximum cord stretch and the maximum g-forces experienced during deceleration by constantly recalculating the system's state. Moreover, we demonstrate how differential equations provide a reliable method for predicting the behavior of non-linear physical systems where simple algebraic solutions are insufficient.

2 Physics Theory and Numerical Method

2.1 Physics Theory

The motion of a bungee jumper is divided into two distinct phases. To model this accurately, we must consider the forces acting on the jumper at every moment.

- **Phase 1: Free Fall** ($y < L$)

Initially, the jumper is in a state of free fall. The only significant force acting on the body is gravity (mg). During this stage, the cord remains slack and exerts no upward force.

- **Phase 2: Elastic Stretching** ($y \geq L$)

Once the jumper falls past the natural length of the cord (L), the cord begins to stretch. According to Hooke's Law, the cord exerts a restorative force proportional to the displacement. The net force becomes a competition between gravity pulling down and the cord pulling up:

$$F_{net} = mg - k(y - L)$$

Where m is the mass, g is gravity, k is the spring constant, and y is the total distance fallen.

2.2 Numerical Method: Euler’s Integration

Since the force changes the moment the cord starts to stretch, standard algebraic kinematics cannot be used. Instead, we utilize a numerical integration method—specifically a loop that updates the system at small time intervals (Δt).

The simulation follows a step-by-step mathematical logic:

1. **Acceleration (a):** We calculate the instantaneous acceleration using Newton’s Second Law ($a = F_{net}/m$).
2. **Velocity (v):** We update the velocity by adding the change caused by acceleration:
 $v_{new} = v_{old} + a \cdot \Delta t$.
3. **Position (y):** We update the position using the new velocity: $y_{new} = y_{old} + v_{new} \cdot \Delta t$.

3 Energy Transformation and Conservation

A key part of our simulation was tracking energy to verify the Law of Conservation of Energy. At the start, the jumper possesses maximum **Gravitational Potential Energy** ($U_g = mgy$). As they fall, this is converted into **Kinetic Energy** ($K = \frac{1}{2}mv^2$).

In Phase 2, Kinetic Energy transfers into **Elastic Potential Energy** ($U_e = \frac{1}{2}k(y - L)^2$). At the point of the **18.9m stretch**, velocity hits zero, meaning energy is fully stored in the cord. Our loop confirmed that the **Total Mechanical Energy** remained constant, proving the numerical method was sound.

4 Simulation Results and Interpretation

Our simulation produced specific data points that help us understand the physical dynamics of the jump:

- **Maximum Cord Stretch (18.9 meters):** The cord reached a maximum stretch of 18.9m. This is the turning point where downward velocity hits zero, ensuring the jumper remains a safe distance from the ground.
- **Maximum G-Forces (1.77g):** The peak force of 1.77g occurred at the bottom of the fall. Since 1g is 9.81 m/s^2 , a result of 1.77 means the jumper felt 1.77 times their body weight, representing a safe and smooth deceleration.

5 Coding Challenges

We faced three main technical challenges during the development of this simulation:

1. **Fixing Cord Logic:** Initially, the model treated the cord like a stiff spring that could “push” the jumper up. We implemented **if/else logic** to ensure the force only applied when $y > L$, keeping the slack phase realistic.
2. **Stopping “Energy Leaks”:** A large time step caused the math to “overshoot,” gaining fake energy. We fixed this by shrinking Δt to **0.01 seconds**, which kept the physics stable and the energy constant.

3. **Visual Layering and Occlusion:** During animation, the jumper sprite would disappear behind the building graphic. We adjusted the **Z-order** to keep the jumper on the top layer and slightly offset the horizontal position for full visibility.

6 Conclusion

The simulation successfully demonstrated the dynamics of a bungee jump. By breaking the motion into phases and using a **0.01s** time step, we achieved a realistic model with an **18.9m stretch** and **1.77g peak**. This project highlights the necessity of numerical methods for solving non-linear, real-world physics problems.

7 Project Resources

The complete source code and presentation for this model can be accessed via the following links:

- **GitHub Repository:** The Bungee Jumper Dynamics Model
- **Presentation Slides:** Overleaf Beamer Slides

8 Simulation Screenshots

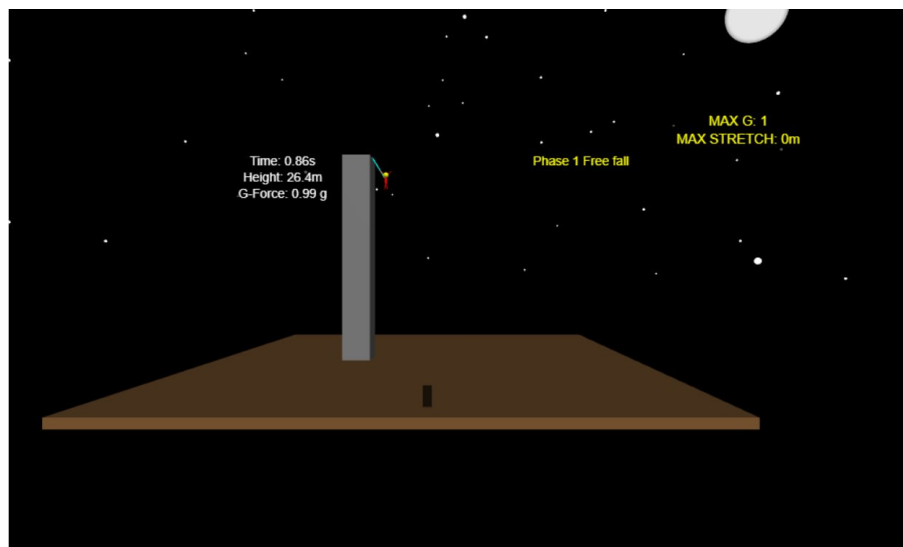


Figure 1: Visual representation of the jumper in free fall.

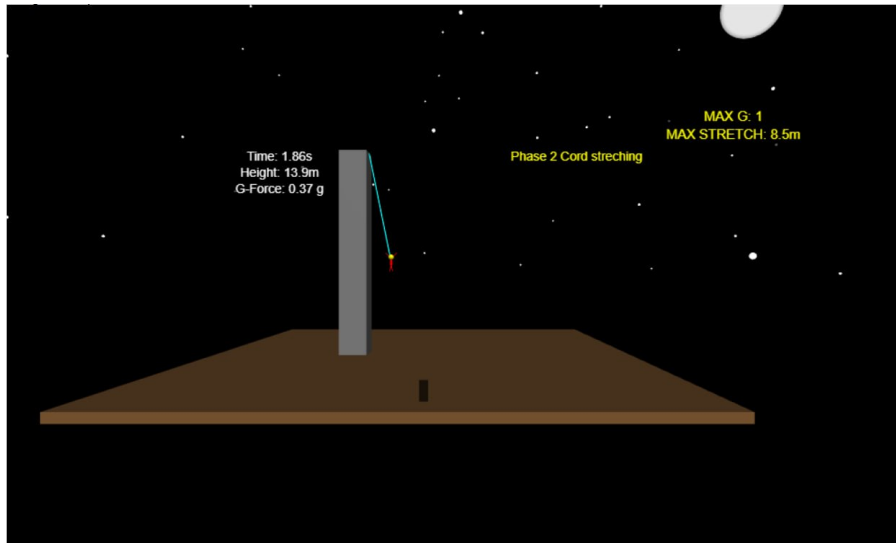


Figure 2: Visual representation of the cord stretching at maximum displacement.

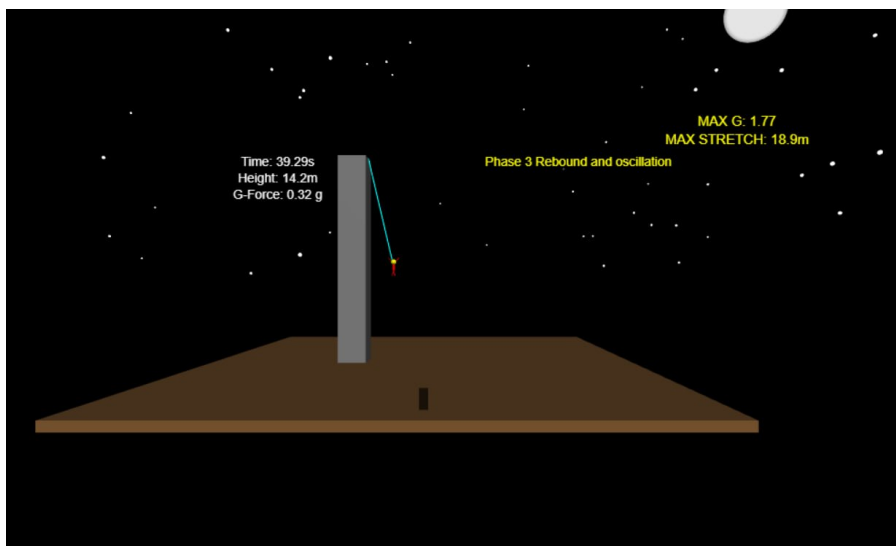


Figure 3: Visual representation of the rebounding and oscillation phase.